UJIAN TENGAH SEMESTER GANJIL
TAHUN AKADEMIK 2025/2026

Mata Kuliah : MATEMATIKA 1

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PRODI D4 TEKNIK INFORMATIKA

UNIVERSITAS HARKAT NEGERI

TAHUN AJARAN 2025/2026

SOAL NO. 1 TEOREMA BAYES

(A) dan (B)

A = 0.4348

B = 0.3067

```
>> tingkat_1 = 50

tingkat_1 =

    50

>> tingkat_2 = 15

tingkat_2 =

    15

>> tingkat_3 = 10

tingkat_3 =

    10

>> total = tingkat_1 + tingkat_2 + tingkat_3

total =

    75
```

```
>> A_tingkat1 = 10

A_tingkat1 =

    10

>> A_tingkat2 = 8

A_tingkat2 =

     8

>> A_tingkat3 = 5

A_tingkat3 =

     5

>> total_A = A_tingkat1 + A_tingkat2 + A_tingkat3

total_A =

    23
```

```
>> peluang_A = total_A / total

peluang_A =

    0.3067

>> peluang_tingkat1_A = (A_tingkat1 / tingkat_1) * (tingkat_1 / total) / peluang_A

peluang_tingkat1_A =

    0.4348

>>
```

SOAL NO. 2 FUNGSI KOMPOSISI

(A)

```
>> syms x
>> f = 2*x - 2

f =

2*x - 2

>> g = 3*x^2 + 10

g =

3*x^2 + 10

>> h = 2*x -1

h =

2*x - 1

>> compose = compose(g, compose(f,h))

compose =

3*(4*x - 4)^2 + 10

>>
```

(B)

```
>> syms x
>> f = 2*x -2

f =

2*x - 2

>> g = 3*x^2 + 10

g =

3*x^2 + 10

>> h = 2*x - 1

h =

2*x - 1

>> compose = compose(f, compose(g,h))

compose =

6*(2*x - 1)^2 + 18

>>
```

SOAL NO 3 FUNGSI INVERS

(A) DAN (B)

```
>> syms x
>> f = 2*x^2 + 4*x + 16

f =

2*x^2 + 4*x + 16

>> finverse (f)

ans =

(2^(1/2)*(x - 14)^(1/2))/2 - 1

>> g = 8*x^2-4*x

g =

8*x^2 - 4*x

>> finverse(f)

ans =

(2^(1/2)*(x - 14)^(1/2))/2 - 1

>>
```

SOAL NO. 4 TURUNAN

(A)

```
>> syms x
>> f = 3*cos(x)

f =

3*cos(x)

>> diff (f)

ans =

-3*sin(x)

>>
```

(B)

```
>> syms x
>> f = 3*x^2 + 5*x + 8

f =

3*x^2 + 5*x + 8

>> diff(f)

ans =

6*x + 5

>>
```

SOAL NO. 5 INTEGRAL

(A) DAN (B)

```
>> syms x
>> f =  4*x^3 + 2*x

f =

4*x^3 + 2*x

>> int(f)

ans =

x^4 + x^2

>> f = 4*x^3 + 5*x

f =

4*x^3 + 5*x

>> int(f, 0, 4)

ans =

296

>>
```